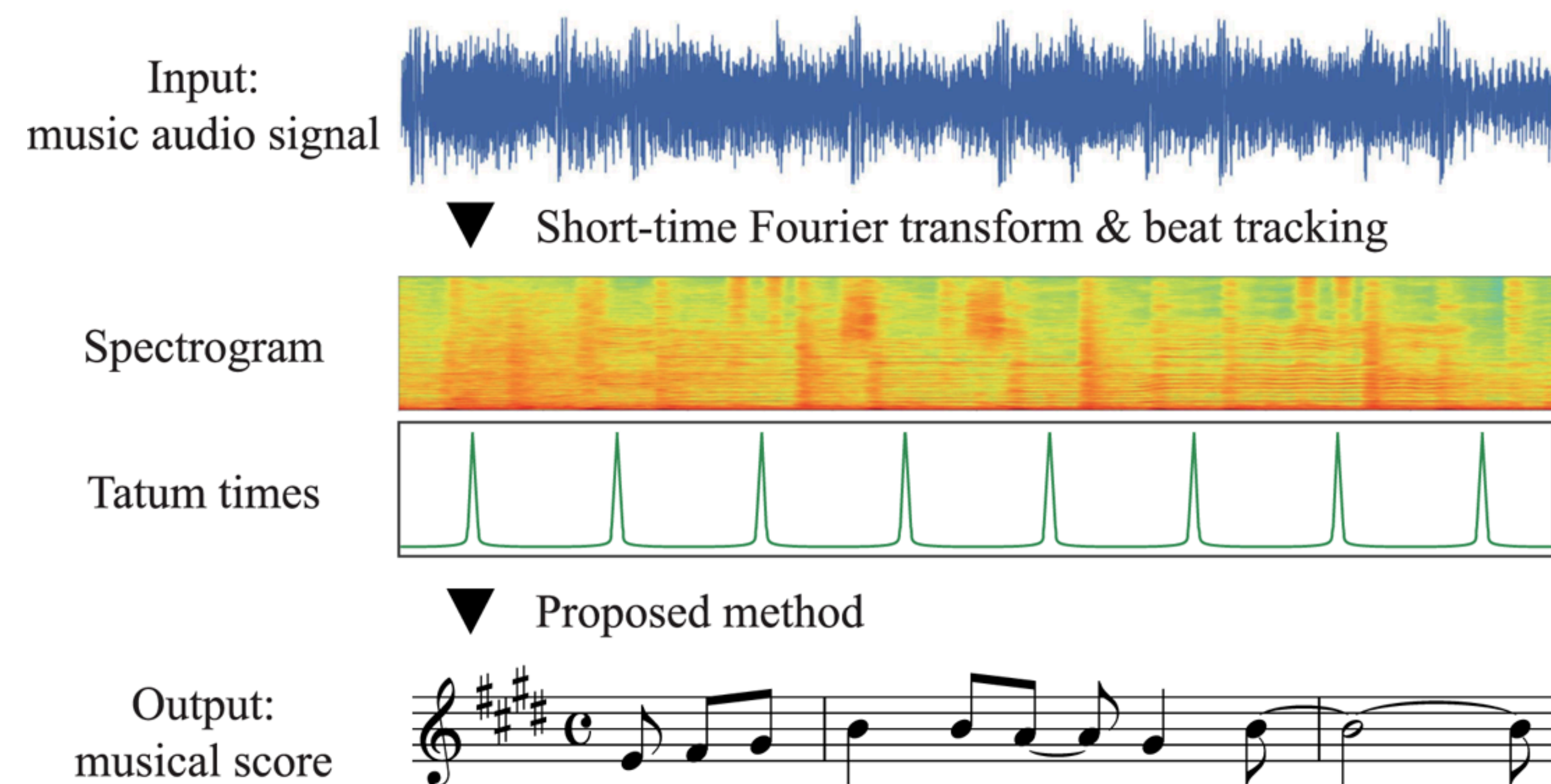


What is AMT?

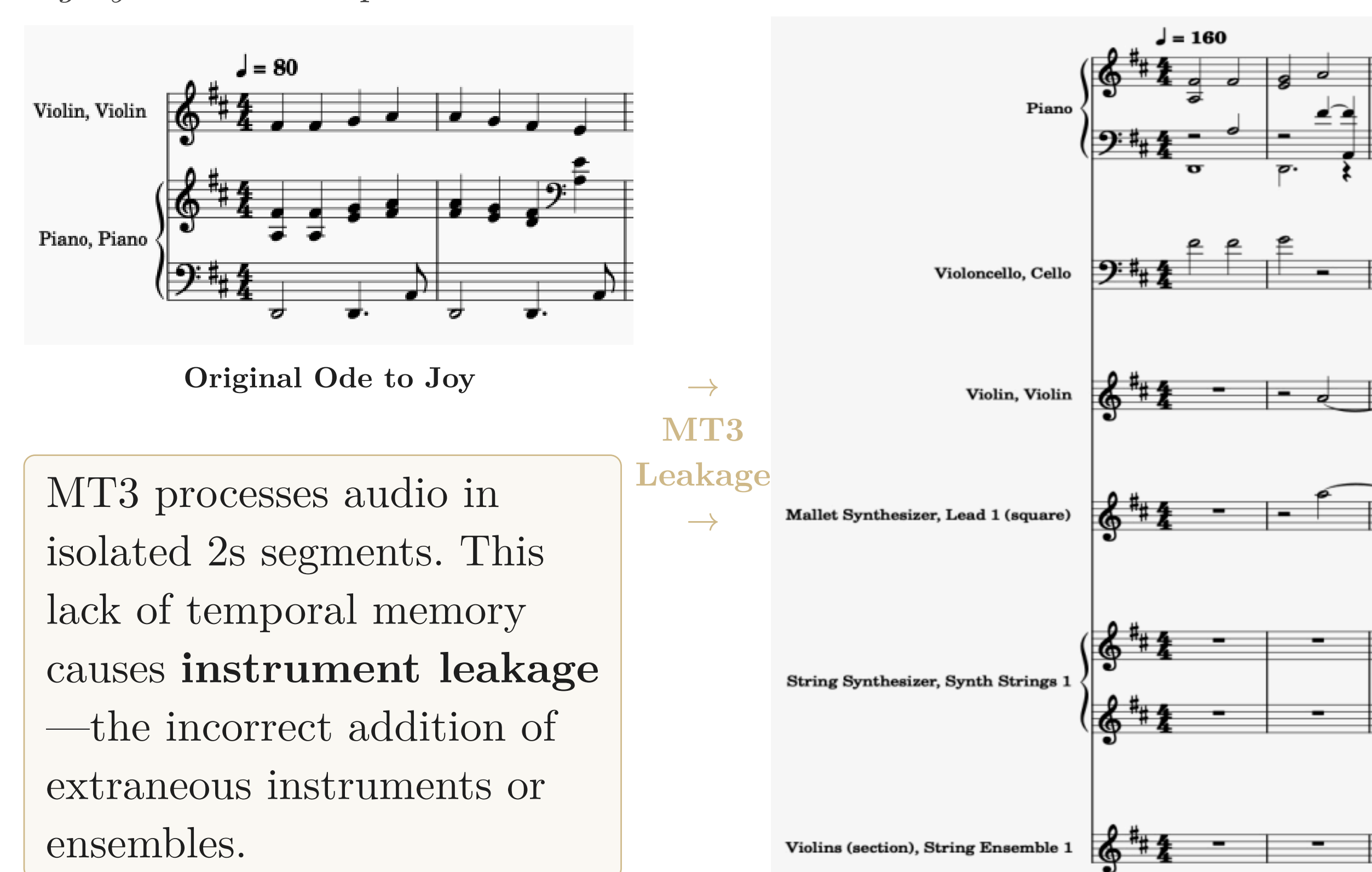
Automatic Music Transcription (AMT) is the computational process of converting raw audio signals into symbolic musical representations like MIDI or digital sheet music.



Existing Model Outputs



Although transcriptions differ visually from the original, the audio remains highly accurate despite minor localized errors.



Original Ode to Joy

→ MT3 Leakage →

MT3 processes audio in isolated 2s segments. This lack of temporal memory causes **instrument leakage**—the incorrect addition of extraneous instruments or ensembles.

Acknowledgements

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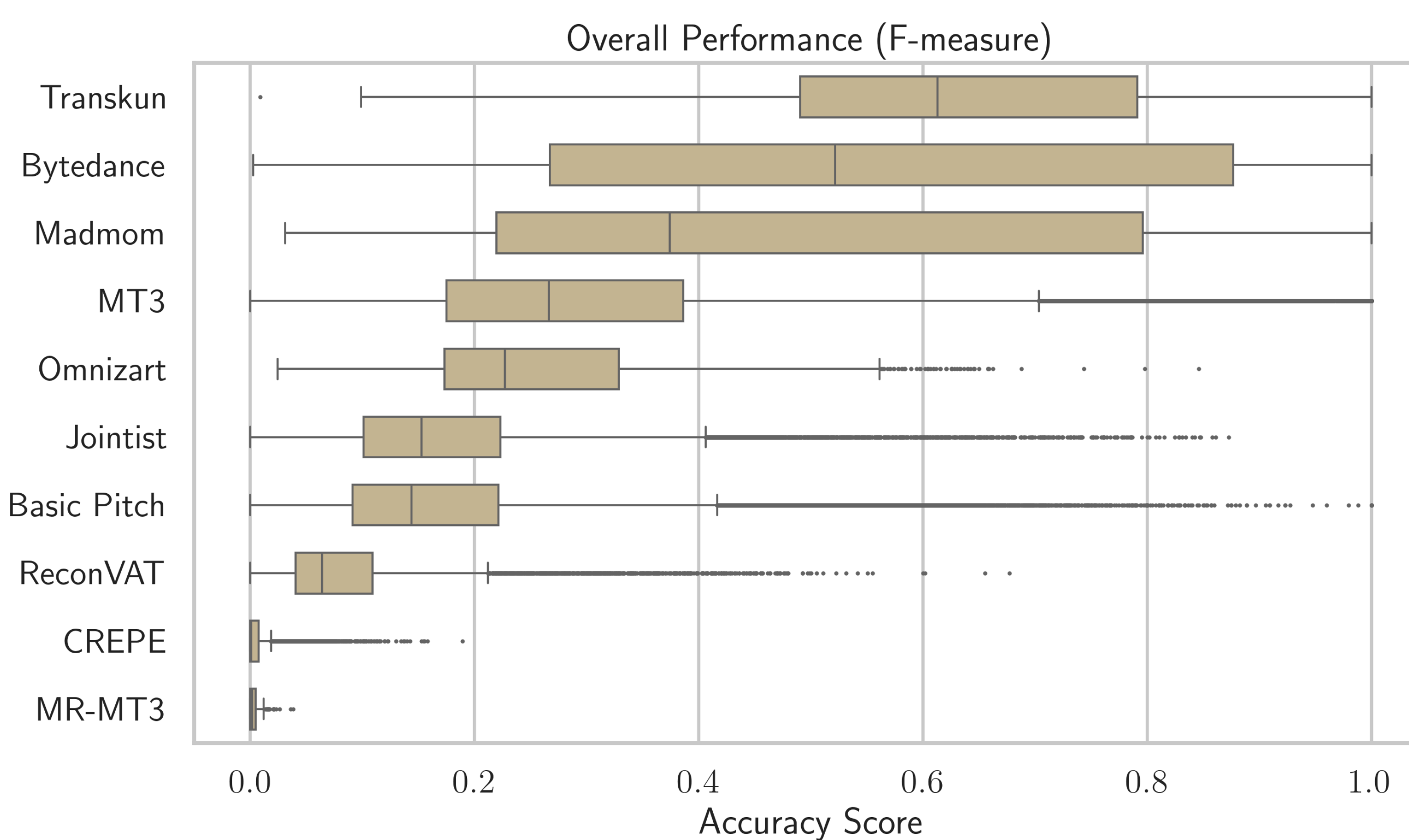


Sources

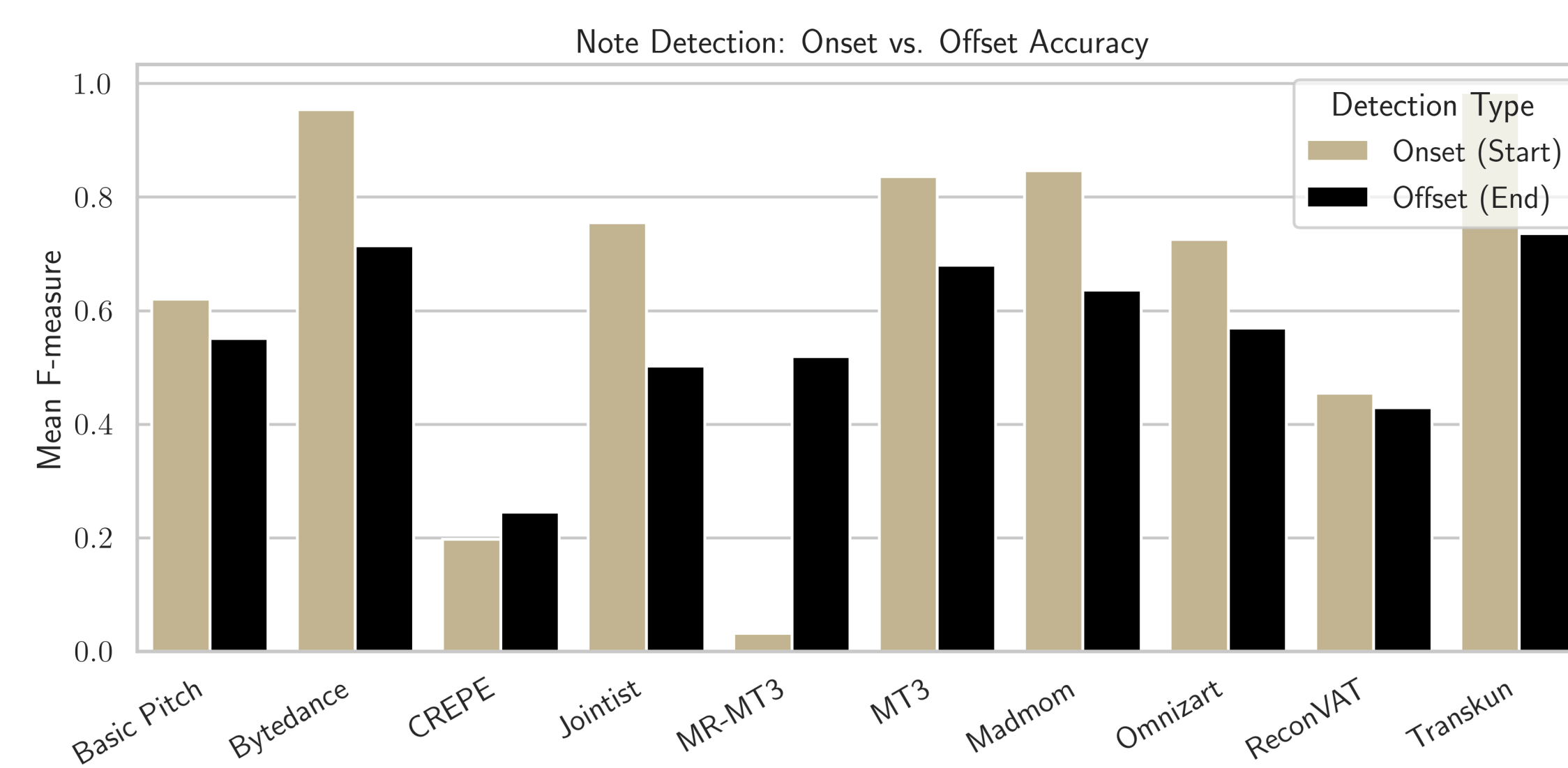
Model Metrics & Evaluation

Comprehensive Performance Benchmarking

Model	Mean F	Std Dev	Onset F	Runtime
Transkun	0.636	0.181	0.985	47.0s
Bytedance	0.570	0.308	0.954	47.6s
Madmom	0.480	0.312	0.846	152.0s
MT3	0.303	0.184	0.836	37.7s
Omnizart	0.265	0.125	0.726	31.1s
Jointist	0.175	0.107	0.755	37.8s
Basic Pitch	0.173	0.119	0.621	27.2s
ReconVAT	0.094	0.087	0.455	7.8s
CREPE	0.008	0.017	0.198	34.0s
MR-MT3	0.003	0.005	0.032	177.6s

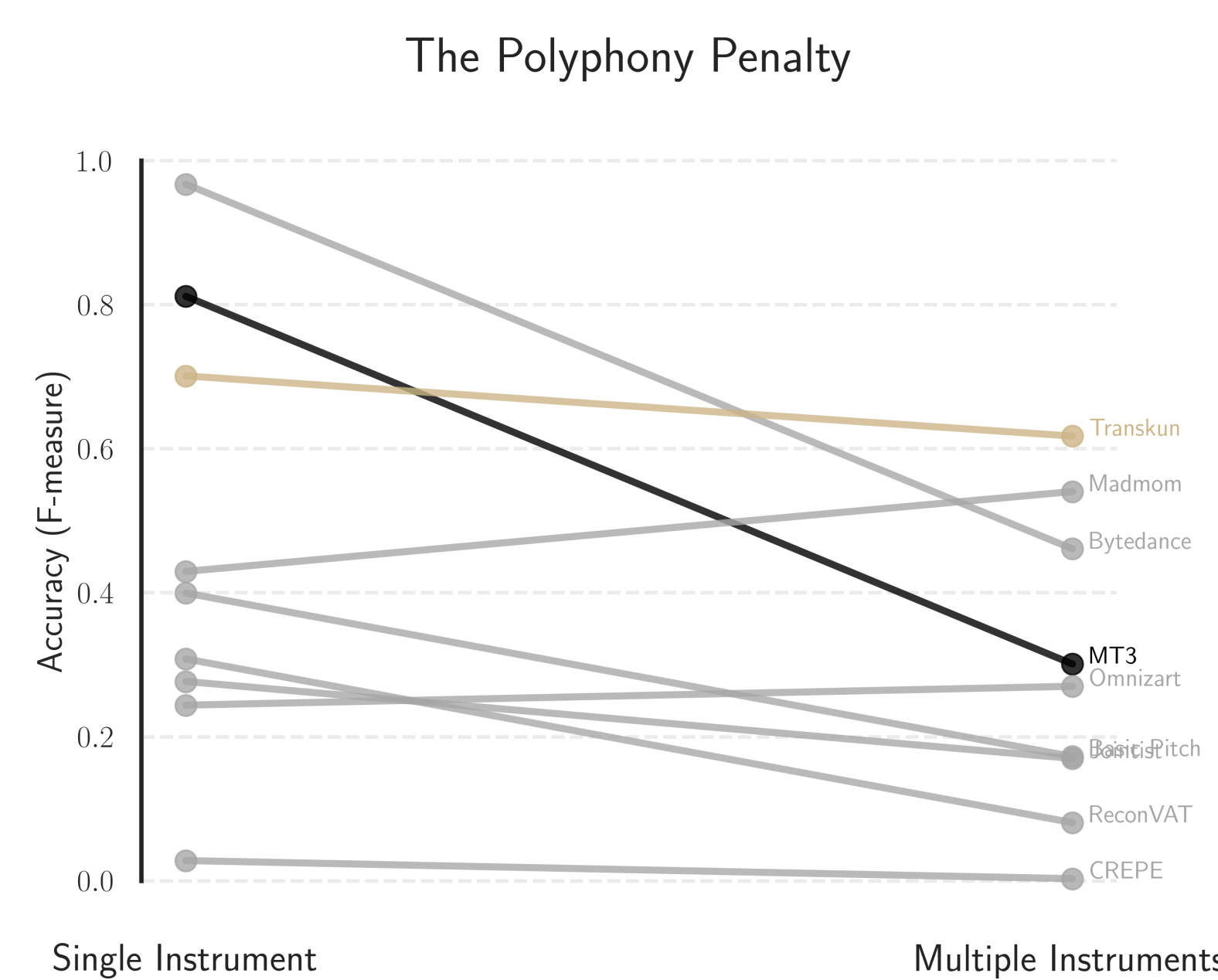


Overall accuracy distribution.



Note Detection Error Gap

Analysis: While Onset detection (Fig 3) is robust across top models, the dramatic drop in Offset accuracy and performance in Polyphonic scenarios (Fig 2) highlights the current limitations in temporal modeling.

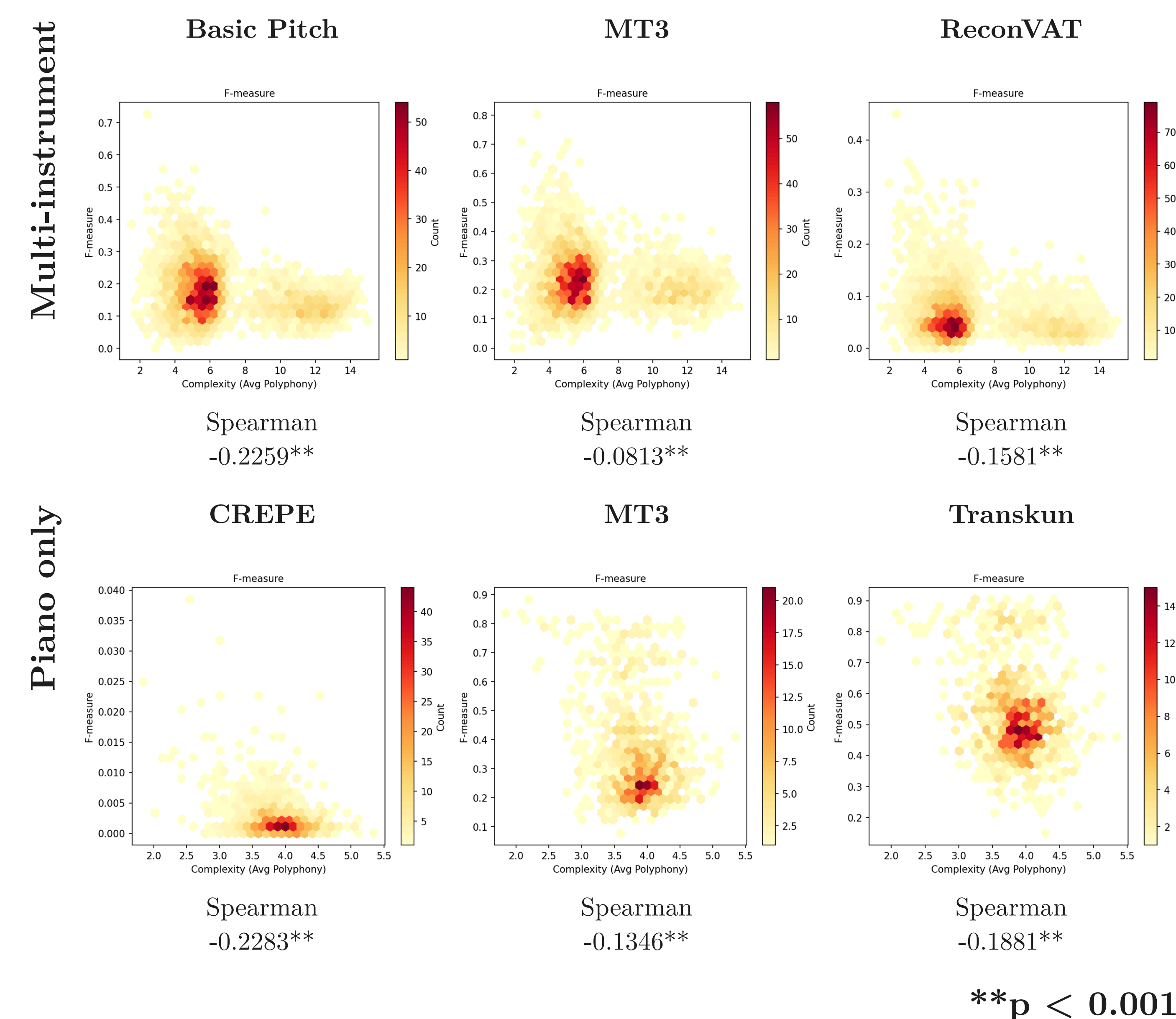


Polyphony Penalty

The Complexity Bottleneck

In reference to the evaluation framework for music transcription models and in order to further improve raw performance, it is relevant to objectively measure the complexity of a piece.

- These “black-boxed” models make it difficult to interpret how results are derived.
- We focus on a subdivision known as **polyphony** and its metric: measure of the average number of notes played simultaneously.
 - We propose that the more complex a piece is, the more difficult it is for models to transcribe.
 - Correlation between specific metrics and accuracy gives insights into features impacting model performance.



Mechanistic Interpretability

What is interpretability?

- Current models are “black boxes”
- We don’t know what they “learn”
- Errors are difficult to isolate and fix
- The “overfitting” problem

Our goal: Build an interpretable AMT model approaching SoTA accuracy with higher transparency. Look inside to see why it does what it does.

$$\begin{matrix} \mathbf{V} \\ \text{(Spectrogram)} \end{matrix} = \begin{matrix} \mathbf{W} \\ \text{(Dictionary)} \end{matrix} \times \begin{matrix} \mathbf{H} \\ \text{(Activations)} \end{matrix}$$

Planned DDS architecture

- Differentiable Dictionary Search
- Dictionary of possible “notes”
- Predicted linear sum of bases